

# Fraud Detection in Financial Transactions

Preliminary Documentation

## Introduction to Artificial Intelligence Project

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**Project Topic:** Fraud Detection

**Dataset:** PaySim Dataset

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# 1 Introduction

Fraud detection is an important application of artificial intelligence in financial systems. Due to the large number of daily transactions, automatic fraud detection models are required to identify suspicious operations efficiently.

The goal of this project is to create a machine learning model that predicts whether a transaction is fraudulent or not based on transaction attributes such as transaction type, amount, balances, and account information.

The task is formulated as a supervised binary classification problem:

$$f(x) = \begin{cases} 1, & \text{fraudulent transaction} \\ 0, & \text{legitimate transaction} \end{cases}$$

## 2 Dataset Description

The project uses the **PaySim Synthetic Financial Dataset** available on Kaggle:

<https://www.kaggle.com/datasets/ealaxi/paysim1>

The dataset simulates mobile money transactions and contains both legitimate and fraudulent operations.

### 2.1 Dataset Statistics

- Total transactions: **6,362,620**
- Fraudulent transactions: **8,213**
- Number of features: **11**
- Fraud ratio: **0.129%**

The dataset is highly imbalanced because fraudulent transactions represent only a very small percentage of all observations.

## 2.2 Main Attributes

Table 1: Selected dataset attributes

| Attribute      | Description                         |
|----------------|-------------------------------------|
| step           | Time step of the transaction        |
| type           | Transaction type                    |
| amount         | Transaction amount                  |
| oldbalanceOrig | Sender balance before transaction   |
| newbalanceOrig | Sender balance after transaction    |
| oldbalanceDest | Receiver balance before transaction |
| newbalanceDest | Receiver balance after transaction  |
| isFraud        | Fraud label                         |

## 3 Planned Data Preprocessing

The following preprocessing steps are planned:

- handling missing or inconsistent values,
- encoding categorical variables,
- feature scaling where necessary,
- feature engineering using balance differences,
- handling class imbalance using oversampling or class weighting,
- splitting the dataset into training and testing sets.

## 4 Algorithms Planned for Use

Several machine learning algorithms will be tested and compared.

### 4.1 Logistic Regression

Logistic Regression will be used as a baseline model because it is simple and interpretable.

$$P(y = 1|x) = \frac{1}{1 + e^{-(w^T x + b)}}$$

## 4.2 Decision Tree

Decision Tree models can capture nonlinear decision rules and provide interpretable classification structures.

## 4.3 Random Forest

Random Forest combines multiple decision trees to improve prediction accuracy and reduce overfitting.

## 4.4 Gradient Boosting

Gradient boosting methods such as XGBoost or LightGBM will also be considered because they achieve strong performance on structured tabular data.

# 5 Planned Experiments

The experiments will include:

- exploratory data analysis,
- training baseline models,
- comparison of different algorithms,
- testing methods for handling class imbalance,
- hyperparameter tuning,
- evaluation using multiple performance metrics.

# 6 Methods of Result Visualization

The following visualizations will be prepared:

- class distribution plots,
- transaction type analysis,
- correlation heatmaps,
- confusion matrices,

- ROC curves,
- precision-recall curves,
- feature importance plots.

## 7 Quality Measures

Because the dataset is highly imbalanced, accuracy alone is insufficient. The following metrics will be used.

### 7.1 Precision

Precision measures how many predicted fraud cases are actually fraudulent.

$$Precision = \frac{TP}{TP + FP}$$

### 7.2 Recall

Recall measures how many fraudulent transactions are correctly detected.

$$Recall = \frac{TP}{TP + FN}$$

### 7.3 F1-Score

F1-score combines precision and recall.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

### 7.4 ROC-AUC

ROC-AUC evaluates the model's ability to distinguish between fraudulent and legitimate transactions.

## 8 Tools and Technologies

The project will be implemented in Python using the following tools:

Table 2: Planned tools

| Tool                 | Purpose                     |
|----------------------|-----------------------------|
| Python               | Main programming language   |
| Pandas               | Data preprocessing          |
| NumPy                | Numerical computations      |
| Matplotlib / Seaborn | Data visualization          |
| Scikit-learn         | Machine learning algorithms |
| XGBoost / LightGBM   | Gradient boosting models    |
| GitHub / GitLab      | Version control             |

## 9 Conclusion

This report presents the preliminary assumptions for the fraud detection project. The project focuses on developing and evaluating machine learning models capable of identifying fraudulent financial transactions using the PaySim dataset. The planned work includes preprocessing, handling class imbalance, training multiple classification models, and evaluating them using appropriate quality measures and visualizations.

## References

- [1] PaySim Dataset, <https://www.kaggle.com/datasets/ealaxi/paysim1>
- [2] Scikit-learn Documentation, <https://scikit-learn.org/>